

Common Diseases & Symptoms

Clinical Reference Guide for Differential Diagnosis Support

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1. Respiratory Diseases

1.1 Influenza (Flu)

Influenza is an acute viral respiratory illness caused by influenza A or B viruses. It spreads primarily via respiratory droplets and is highly contagious, particularly in enclosed spaces. Seasonal epidemics occur annually, typically during winter months.

Classic Presentation

- Sudden onset of high fever (38–40°C)
- Severe myalgia and arthralgia
- Dry, persistent cough
- Headache and retro-orbital pain
- Malaise and fatigue lasting 1–2 weeks
- Sore throat and nasal congestion (variable)

■ **EMERGENCY:** Influenza can rapidly progress to viral pneumonia, ARDS, or secondary bacterial pneumonia. Immediate evaluation required if dyspnea, hypoxia (SpO2 < 92%), or altered consciousness develops.

■ High-risk populations: Age >65, pregnancy, immunocompromised states, chronic cardiopulmonary disease. Annual vaccination is the primary prevention strategy.

1.2 Pneumonia

Pneumonia is an infection of the lung parenchyma classified by causative organism (bacterial, viral, fungal, atypical), setting of acquisition (community-acquired, hospital-acquired, ventilator-associated), and radiographic pattern (lobar, broncho, interstitial).

Symptoms by Severity

Feature	Mild (Outpatient)	Moderate	Severe (ICU)
Fever	Low-grade	38–39°C	>39°C or hypothermia
Respiratory Rate	<20/min	20–30/min	>30/min

Feature	Mild (Outpatient)	Moderate	Severe (ICU)
SpO2	>94%	90–94%	<90%
Confusion	Absent	Absent/mild	Present
BP	Normal	Normal	Systolic <90 mmHg

1.3 Asthma

Asthma is a chronic inflammatory airway disease characterised by reversible airflow obstruction, airway hyperresponsiveness, and variable symptoms triggered by allergens, exercise, cold air, or irritants.

- Episodic wheezing (expiratory > inspiratory)
- Chest tightness, particularly at night
- Shortness of breath on exertion
- Persistent dry cough (especially nocturnal)
- Symptoms improve with bronchodilators

2. Cardiovascular Diseases

2.1 Hypertension

Essential hypertension (primary hypertension) accounts for 90–95% of all hypertension cases. It is a major modifiable risk factor for stroke, myocardial infarction, and chronic kidney disease.

Classification	Systolic (mmHg)	Diastolic (mmHg)	Action
Normal	<120	<80	Lifestyle maintenance
Elevated	120–129	<80	Lifestyle modification
Stage 1 HTN	130–139	80–89	Lifestyle ± medication
Stage 2 HTN	≥140	≥90	Medication required
Hypertensive Crisis	≥180	≥120	Emergency evaluation

■ **Hypertensive emergency (BP ≥180/120 with end-organ damage): Activate emergency services immediately. Target BP reduction of no more than 25% in the first hour.**

2.2 Acute Myocardial Infarction (Heart Attack)

AMI results from complete or near-complete occlusion of a coronary artery, most commonly by rupture of an atherosclerotic plaque followed by thrombosis. Irreversible myocardial injury begins within 20–40 minutes of

ischaemia.

- Crushing, pressure-like chest pain radiating to left arm, jaw, or back
- Diaphoresis (cold sweats) and pallor
- Nausea, vomiting, and epigastric discomfort
- Dyspnoea and profound fatigue
- Atypical presentation in women and diabetics: jaw pain, fatigue, or no chest pain
- New onset arrhythmias or syncope

■ **EMERGENCY: Suspected MI — Call emergency services immediately. Time to reperfusion (PCI) < 90 minutes is the key metric. Aspirin 300 mg stat if not contraindicated.**

3. Gastrointestinal Diseases

3.1 Gastroesophageal Reflux Disease (GERD)

GERD is caused by incompetence of the lower oesophageal sphincter, allowing acidic gastric contents to reflux into the oesophagus, causing mucosal irritation and characteristic symptoms.

- Heartburn (burning retrosternal sensation after meals)
- Regurgitation of acid or undigested food
- Dysphagia (with progressive disease)
- Chronic cough, hoarseness, laryngitis (extraoesophageal)
- Worsening when supine or after large meals

3.2 Peptic Ulcer Disease

Peptic ulcers are mucosal defects in the stomach or duodenum caused predominantly by *H. pylori* infection (70–80%) or NSAID use. Gastric acid perpetuates mucosal damage once the protective barrier is breached.

- Epigastric pain: burning or gnawing, often relieved by food (duodenal) or worsened (gastric)
- Nausea, bloating, and early satiety
- Weight loss (gastric ulcer)
- Signs of complication: haematemesis, melaena, severe acute abdominal pain (perforation)

■ **Red flags requiring urgent referral: unintentional weight loss, dysphagia, haematemesis, melaena, persistent vomiting, palpable mass, age >55 with new-onset dyspepsia.**

4. Endocrine Disorders

4.1 Diabetes Mellitus

Feature	Type 1 DM	Type 2 DM
Onset	Childhood/adolescence	Usually adult (>40 yrs)
Mechanism	Autoimmune beta-cell destruction	Insulin resistance + secretory defect
Body Habitus	Normal or thin	Overweight/obese (typically)
Ketosis	Prone (DKA)	Uncommon (HHS risk)
Primary Rx	Insulin therapy	Lifestyle, oral agents, GLP-1, insulin

Classic hyperglycaemic symptoms: polyuria, polydipsia, polyphagia, blurred vision, unintentional weight loss, fatigue, and recurrent infections (particularly candidal).

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